JOHN JAY COLLEGE OF CRIMINAL JUSTICE
The City University of New York
524 West 59th Street, New York, NY, 10019

Syllabus for: MAT/FOS 705
Applied Statistics and Data Analysis for Forensic Science and Forensic Computing

Professor's name: Nicholas Petraco
Laboratory: 5.64 New Building
Contact hours: Thursdays 6:00pm-8:00pm

E-mail address: npetraco@gmail.com
Course website: https://npetraco.github.io/MATFOS705/

Course Description:

Physical evidence comparison has been "put under the microscope" since the 2009 publication of the National Academy of Sciences report "Strengthening Forensic Science in the United Stated: A Path Forward". The implementation of more quantitative and statistically-based standard operating procedures is a foregone conclusion.

The classical "work horse" statistical methods learned in a first year statistics course are known as "frequentist" and "Fisherian" methods. They are extremely useful and have been successfully applied for almost one hundred years. Forensic scientists not familiar with statistical methods of data analysis will be at a severe disadvantage. The purpose of this hands-on course is to acquaint forensic computing and forensic science graduate students with statistical methods that are applicable to evidence they will encounter, as well as, issues and pitfalls to be aware of when applying these methods.

It is not adequate to simply learn the "theory" behind the methodology encountered in this course. It must be applied on real data of practical interest to the forensic science. As such this course will build expertise in the general scientific/statistical computing environment R (http://www.r-project.org/). The course assumes minimal knowledge of numerical computeing and statistical procedures. It is designed to build the student's skill set and confidence in both of these areas. Topics covered will include basic data descriptive tools, graphing, probability theory, discrete/continuous distributions, estimation, hypothesis testing and regression.

Course materials will be posed on the following website:
https://npetraco.github.io/MATFOS705/

Course Learning Goals:

1. Recognize the importance of accuracy and objectivity in collecting/sampling data for applications to the law.

2. Acquire an understanding of the types of data that can be recorded and analyzed for quantitative trace, fire debris, toolmark, spectrochemical and digital evidence analysis.

3. Acquire an understanding of frequentist statistical tools that can be used to analyze collected forensic data.

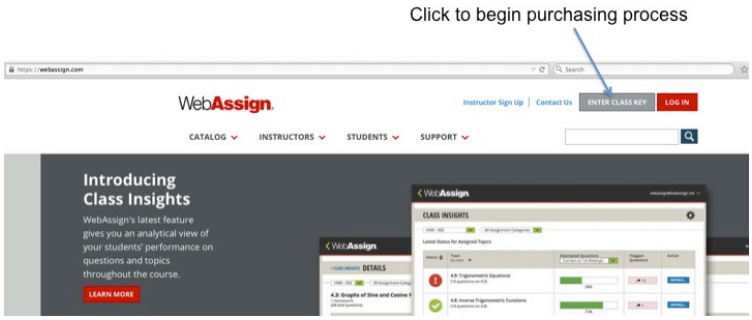
4. Understand the limitations of the frequentist statistical methods used for data analytics and how not to misrepresent the capabilities of these methods to the courts or clients (ethics).

5. Obtain skill with the general computing/statistical software **R** (http://www.r-project.org/).

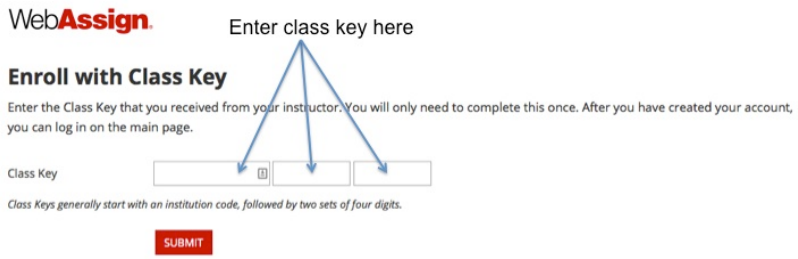
6. Develop oral and written communication skill as to how to present the results of sophisticated quantitative analysis to officers of the course and lay juries, in terms that are understandable to them.

Required Electronic Resources:

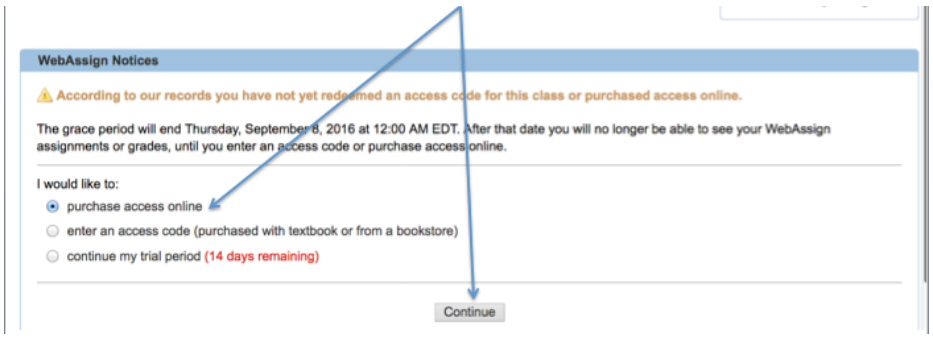
- Webassign: Can be purchased at:
  - **https://webassign.com/**
  - In order to purchase click on "**Enter Class Key**":



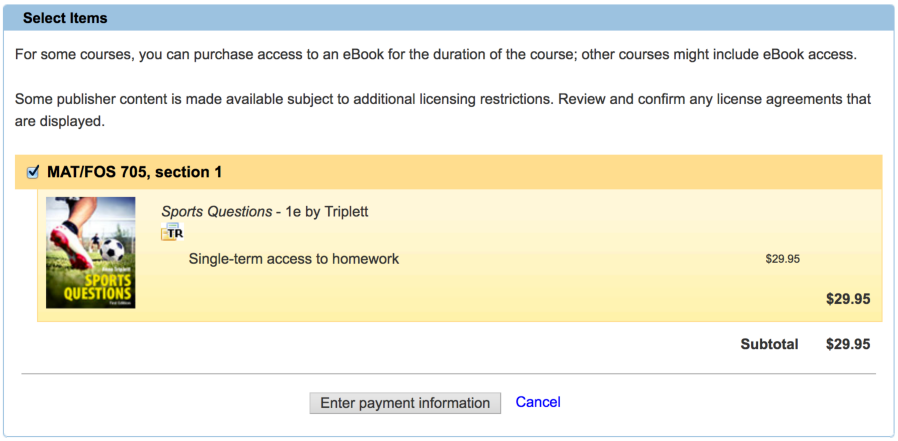
- **You should see a place to enter the class key**:
  - Class Key: **jjay.cuny 0270 3824**



- After logging in/creating-account, select your class and the website will prompt you to purchase the required materials:



- Purchase "MAT/FOS 705 section 1", which should be ~$29.95:



Required textbook:
Statistics and Data with R: An applied approach through examples
Cohen & Cohen
ISBN-10: 0470758058

Suggest/Reference textbooks:
Introduction to Data Analysis with R for Forensic Scientists
Curran
ISBN-10: 1420088262

The R Book
Crawley
ISBN-10: 8126569719

Grading:

The grades for this course are based on home works (100%).

Course lecture/laboratory calendar:

Lecture	Date	Topic
1	Jan 31	Introduction, Terminology, Basic Graphing
1		Introduction To R
2	Feb 7	Summarizing Data, Probability
2		Discrete Distributions
3	Feb 14*	Discrete Distributions
4	Feb 21	Continuous Distributions
5	Feb 28	Sampling Distributions for Important Summary Statistics
6	Mar 7	Point Estimation
7	Mar 14	Interval Estimation
8	Mar 21	Single Sample Hypothesis Tests
9	Mar 28	Sample Size Estimation for Single Sample Hyp. Tests
9		Two Sample Hypothesis Tests
10	Apr 4	Two Sample Hypothesis Tests
10		Sample Size Estimation for Two Sample Hyp. Tests
11	Apr 11	One-Way Analysis of Variance
12	Apr 18	One-Way Analysis of Variance
	Apr 25*	No Class
13	May 2	Two-Way Analysis of Variance
14	May 9	Regression

College wide policies for undergraduate courses (see the *Undergraduate Bulletin*, Chapter IV Academic Standards)

A. **Incomplete Grade Policy**

B. **Extra Work During the Semester**

C. **Americans with Disabilities Act (ADA) Policies**

"Qualified students with disabilities will be provided reasonable academic accommodations if determined eligible by the Office of Accessibility Services (OAS). Prior to granting disability accommodations in this course, the instructor must receive written verification of a student's eligibility from the OAS which is located at L66 in the new building (212-237-8031). It is the student's responsibility to initiate contact with the office and to follow the established procedures for having the accommodation notice sent to the instructor."

Source: *Reasonable Accommodations: A Faculty Guide to Teaching College*

Students with Disabilities, 4th ed., City University of New York, p.3. (http://www.jjay.cuny.edu/studentlife/Reasonable_Accommodations.pdf)

Statement of the College Policy on Plagiarism

Plagiarism is the presentation of someone else's ideas, words, or artistic, scientific, or technical work as one's own creation. Using the ideas or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations require citations to the original source.

Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism.

It is the student's responsibility to recognize the difference between statements that are common knowledge (which do not require documentation) and restatements of the ideas of others. Paraphrase, summary, and direct quotation are acceptable forms of restatement, as long as the source is cited.

Students who are unsure how and when to provide documentation are advised to consult with their instructors. The Library has free guides designed to help students with problems of documentation. *(John Jay College of Criminal Justice Undergraduate Bulletin,* http://www.jjay.cuny.edu/academics/654.php , see Chapter IV Academic Standards)